CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 2: Virtualization Technologies & Hypervisor Types

Course: Open Elective III [23OECS31]
Category: Assignments | Assignment #2 | Academic Year: 2026-27

---

1. Overview & Objective:

Hardware virtualization, hypervisor architectures, and memory virtualization techniques.

2. Assignment Problems & Instructions:

1. Compare Type-1 bare-metal hypervisors ESXi, Xen with Type-2 hosted hypervisors VirtualBox .
2. Explain Intel VT-x and AMD-V hardware-assisted CPU virtualization rings Ring -1 .
3. Describe memory overcommitment, memory ballooning, and transparent page sharing.
4. Analyze network virtualization through Virtual Switches vSwitch and VXLAN overlay networks.

Submission Guidelines & Verification:

This document verifies end-to-end PDF storage, preview, and download for CampusNotes.
Administrators can safely edit metadata, publish/unpublish, or replace this document.